1. The number of moles in 22.4 L of a gas at STP is

A) 0.5

B) 1

C) 2

D) 22.4

Answer: B

---

2. The SI unit of amount of substance is

A) Gram

B) Mole

C) Kilogram

D) Liter

Answer: B

---

3. Atomic number represents

A) Protons

B) Neutrons

C) Electrons

D) Mass

Answer: A

---

4. Isotopes have same

A) Mass number

B) Atomic number

C) Neutrons

D) Protons + Neutrons

Answer: B

---

5. The shape of s orbital is

A) Dumbbell

B) Spherical

C) Linear

D) Planar

Answer: B

---

6. Maximum electrons in p orbital is

A) 2

B) 6

C) 10

D) 14

Answer: B

---

7. Pauli exclusion principle states

A) Same energy

B) Opposite spins

C) Same spins

D) None

Answer: B

---

8. Aufbau principle is based on

A) Energy

B) Mass

C) Charge

D) Spin

Answer: A

---

9. The most electronegative element is

A) Oxygen

B) Nitrogen

C) Fluorine

D) Chlorine

Answer: C

10. Ionization energy increases across period due to

A) Decrease in size

B) Increase in size

C) Shielding

D) None

Answer: A

---

11. Covalent bond involves

A) Transfer

B) Sharing

C) Loss

D) Gain

Answer: B

---

12. Ionic bond forms by

A) Sharing

B) Transfer

C) Hybridization

D) Overlap

Answer: B

---

13. VSEPR theory predicts

A) Bond energy

B) Shape

C) Mass

D) Charge

Answer: B

---

14. Hybridization of methane is

A) sp

B) sp^2

C) sp^3

D) sp^3d

Answer: C

---

15. The bond angle in methane is

A) 90°

B) 109.5°

C) 120°

D) 180°

Answer: B

---

16. pH of neutral solution is

A) 0

B) 7

C) 14

D) 1

Answer: B

---

17. Strong acid has

A) Low ionization

B) High ionization

C) Neutral

D) Weak bond

Answer: B

---

18. Buffer solution resists

A) Temperature

B) Pressure

C) pH change

D) Volume

Answer: C

---

19. Oxidation is

A) Gain of electrons

B) Loss of electrons

C) Gain of protons

D) Loss of neutrons

Answer: B

---

20. Reduction is

A) Loss of electrons

B) Gain of electrons

C) Loss of protons

D) Gain of neutrons

Answer: B

---

21. Catalyst increases

A) Energy

B) Rate

C) Temperature

D) Pressure

Answer: B

---

22. Activation energy is

A) Initial energy

B) Minimum energy

C) Maximum energy

D) Total energy

Answer: B

---

23. Rate of reaction depends on

A) Temperature

B) Concentration

C) Catalyst

D) All

Answer: D

---

24. Equilibrium constant depends on

A) Pressure

B) Temperature

C) Concentration

D) Catalyst

Answer: B

---

25. Le Chatelier principle deals with

A) Equilibrium shift

B) Energy

C) Bond

D) Mass

Answer: A

---

26. Electrolysis involves

A) Chemical energy

B) Electrical energy

C) Heat

D) Light

Answer: B

---

27. Faraday's law relates to

A) Mass

B) Charge

C) Energy

D) Volume

Answer: B

---

28. The unit of conductance is

A) Ohm

B) Siemens

C) Volt

D) Ampere

Answer: B

---

29. Oxidation number of oxygen is usually

A) +2

B) -2

C) 0

D) +1

Answer: B

---

30. Alkali metals belong to group

A) 1

B) 2

C) 17

D) 18

Answer: A

---

31. Halogens belong to

A) Group 1

B) Group 2

C) Group 17

D) Group 18

Answer: C

---

32. Noble gases are

A) Reactive

B) Inert

C) Metals

D) Liquids

Answer: B

---

33. Carbon forms

A) 2 bonds

B) 3 bonds

C) 4 bonds

D) 5 bonds

Answer: C

---

34. Functional group in alcohol is

A) -COOH

B) -OH

C) -CHO

D) $-NH_2$

Answer: B

---

35. Functional group in aldehyde is

A) -CHO

B) -OH

C) -COOH

D) $-NH_2$

Answer: A

---

36. Functional group in carboxylic acid is

A) -CHO

B) -OH

C) -COOH

D) $-NH_2$

Answer: C

---

37. Saturated hydrocarbons are

A) Alkenes

B) Alkynes

C) Alkanes

D) Aromatics

Answer: C

---

38. Ethene has

A) Single bond

B) Double bond

C) Triple bond

D) Ionic bond

Answer: B

---

39. Benzene is

A) Aliphatic

B) Aromatic

C) Saturated

D) Alkane

Answer: B

---

40. Isomerism is

A) Same formula, different structure

B) Same structure

C) Same mass

D) Same atoms

Answer: A

41. Polymerization forms

A) Small molecules

B) Large molecules

C) Ions

D) Atoms

Answer: B

---

42. Protein is made of

A) Sugars

B) Amino acids

C) Fatty acids

D) Nucleotides

Answer: B

---

43. Enzymes are

A) Lipids

B) Proteins

C) Carbohydrates

D) Vitamins

Answer: B

---

44. DNA contains

A) Sugar

B) Phosphate

C) Base

D) All

Answer: D

---

45. The gas evolved in photosynthesis is

A) CO_2

B) O_2

C) N_2

D) H_2

Answer: B

---

46. The gas used in respiration is

A) CO_2

B) O_2

C) N_2

D) H_2

Answer: B

---

47. Hard water contains

A) Na^+

B) Ca^{2+}

C) K^+

D) H^+

Answer: B

---

48. Temporary hardness is due to

A) Sulfates

B) Bicarbonates

C) Chlorides

D) Nitrates

Answer: B

---

49. Permanent hardness is due to

A) Bicarbonates

B) Sulfates

C) CO_2

D) H_2

Answer: B

---

50. Greenhouse gas is

A) O_2

B) CO_2

C) N_2

D) H_2

Answer: B

---

51. Ozone layer protects from

A) IR rays

B) UV rays

C) Visible light

D) Heat

Answer: B

---

52. The p-block elements are in

A) Left

B) Right

C) Middle

D) Bottom

Answer: B

---

53. Transition elements are

A) s-block

B) p-block

C) d-block

D) f-block

Answer: C

---

54. Lanthanides belong to

A) s-block

B) p-block

C) d-block

D) f-block

Answer: D

---

55. The strongest intermolecular force is

A) Hydrogen bond

B) Dipole

C) London force

D) Ionic

Answer: A

---

56. Boiling point increases with

A) Weak forces

B) Strong forces

C) Low mass

D) None

Answer: B

---

57. Molarity is

A) mol/L

B) mol/kg

C) g/L

D) g/kg

Answer: A

---

58. Molality is

A) mol/L

B) mol/kg

C) g/L

D) g/kg

Answer: B

---

59. Colligative properties depend on

A) Nature

B) Number

C) Size

D) Shape

Answer: B

---

60. Raoult's law applies to

A) Ideal solutions

B) Real solutions

C) Gases

D) Solids

Answer: A